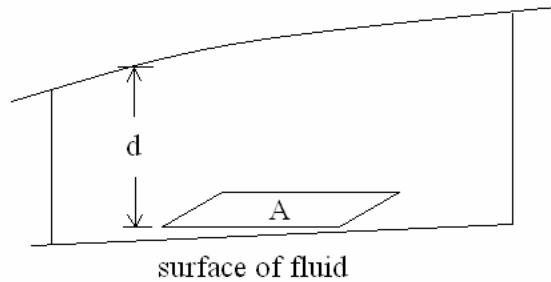


§8-3 Applications to Physics and Engineering

Homework : 1,11,27,34

(I) Hydrostatic Pressure and Force



$$p = \frac{F}{A} = \frac{mg}{A} = \frac{\rho A d}{A} = \rho g d.$$

F : force m : mass g : gravity acceleration.

ρ : density of a fluid in metric system.

δ : density of a fluid in ft-pound system.

Water density : $1000 \frac{\text{kg}}{\text{m}^3}$ or $62.5 \text{ pd}/\text{ft}^3$ (含重力)

$$(1) \quad P(\text{pressure}) = \begin{cases} \rho g d & (\text{metric system}) \\ \delta d & (\text{ft-pound system}). \end{cases}$$

Force against a vertical plate or wall or dam.

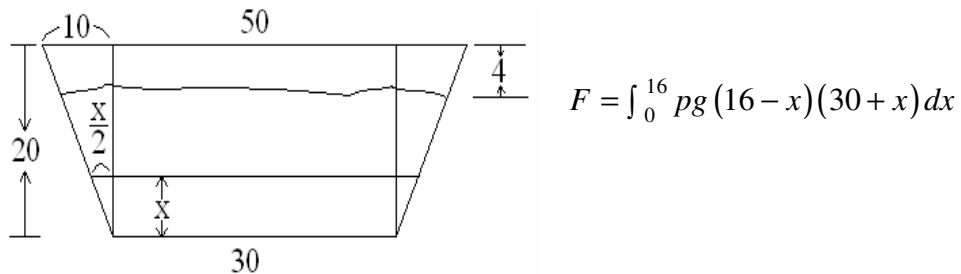
(2)



$$F = \int p dA \quad (p, A \text{ 隨位置而變})$$

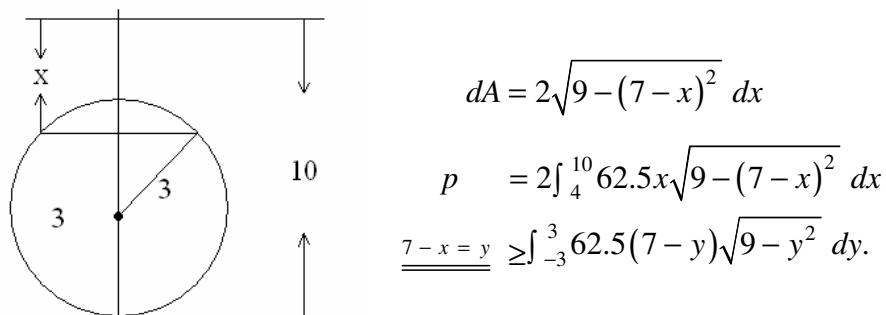
Example 1 : A dam has the shape of the trapezoid. The height is 20m, and width is 50m at the top and 30m at the bottom. Find the force on the dam due to hydrostatic pressure if the water level is 4 m from the top of the dam.

Solution :



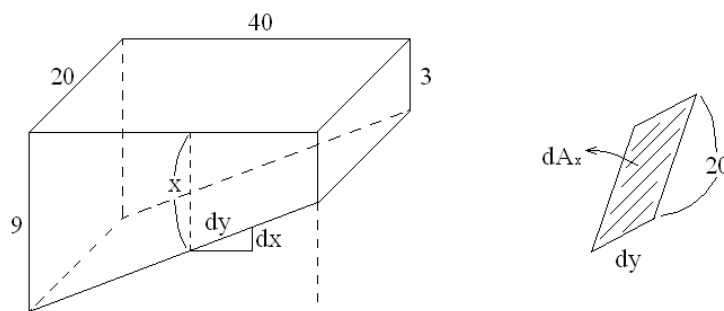
Example 2 : Find the hydro static force on one end of a cylindrical drum with radius 3 ft if the drum is submerged in water 10 ft deep.

Solution :



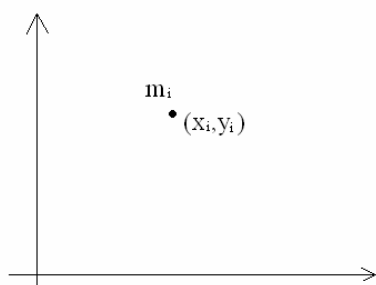
Example 3 : Force against an inclined bottom.

Solution :



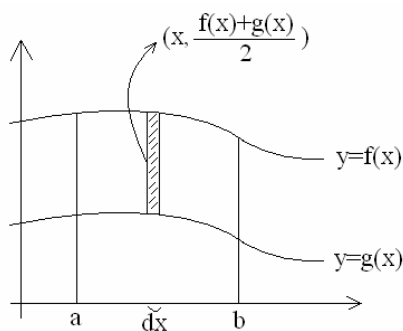
$$\begin{aligned}
& \int \frac{9}{3} \rho dA_x \\
&= \int \frac{9}{3} \delta dA_x \\
&= \int \frac{9}{3} \delta x 20 dy \\
&\quad \frac{dy}{dx} = \frac{\sqrt{1636}}{6} = \frac{\sqrt{409}}{3} \quad \Rightarrow \quad dy = \frac{\sqrt{409}}{3} dx \\
&= \int \frac{9}{3} \frac{20\sqrt{409}}{3} \delta x dx.
\end{aligned}$$

Center $(\bar{x}, \bar{y})$ of a discrete system.



$$\begin{aligned}
M_x &= \sum_{i=1}^n m_i y_i \\
M_y &= \sum_{i=1}^n m_i x_i \\
\Rightarrow \bar{x} &= \frac{M_y}{\sum_{i=1}^n m_i}, \quad \bar{y} = \frac{M_x}{\sum_{i=1}^n m_i}
\end{aligned}$$

Center $(\bar{x}, \bar{y})$ of a continuous system with uniform density ρ .



$$\left(x, \frac{f(x) + g(x)}{2} \right): \text{center of infinitesimal strip}$$

at x (the shaded part on the right.).

$$M_x = \int_a^b \frac{(f(x) + g(x))}{2} \cdot \rho (f(x) + g(x)) dx = \rho \int_a^b \frac{(f^2(x) + g^2(x))}{2} dx.$$

$$M_y = \int_a^b x \cdot \rho (f(x) + g(x)) dx = \rho \int_a^b x \cdot (f(x) + g(x)) dx.$$

$$\Rightarrow \bar{y} = \frac{\frac{1}{2} \int_a^b (f^2(x) - g^2(x)) dx}{\int_a^b (f(x) - g(x)) dx}$$

$$\bar{x} = \frac{\int_a^b x(f(x) - g(x)) dx}{\int_a^b (f(x) - g(x)) dx}$$

Example 1 : Find the center of a semicircular plate of radius r .

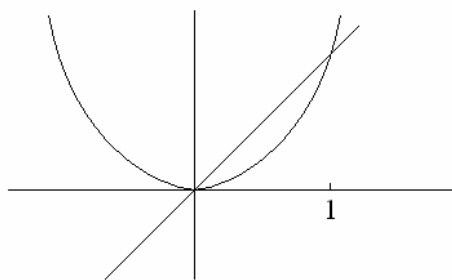
Solution :

$$\Rightarrow \bar{y} = \frac{\frac{1}{2} \int_{-r}^r (\sqrt{r^2 - x^2})^2 dx}{\frac{1}{2} \pi r^2} = \frac{\int_{-r}^r (r^2 - x^2) dx}{\pi r^2}$$

$$= \frac{1}{\pi r^2} \frac{1}{2} (r^2 x - \frac{x^3}{3}) \Big|_{-r}^r = \frac{4r}{3\pi}$$

Example 2 : Find the centroid of the region bounded by $y = x$ and $y = x^2$.

Solution :



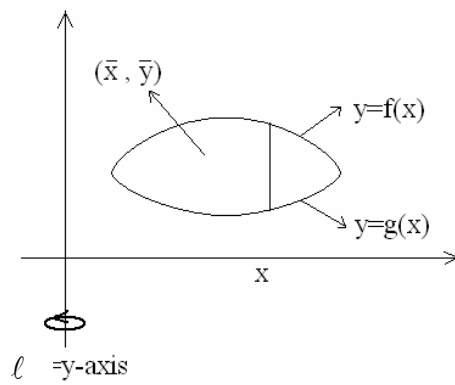
$$A = \int_0^1 (x - x^2) dx = \frac{1}{6}$$

$$M_y = \int_0^1 \frac{1}{2} (x - x^2) dx = \frac{1}{15}$$

$$M_x = \int_0^1 x(x - x^2) dx = \frac{1}{12}$$

$$\Rightarrow \bar{x} = \frac{1}{2}, \quad \bar{y} = \frac{2}{5}.$$

Theorem of Pappu :



$$\begin{aligned}
 V &= \int_a^b 2\pi x (f(x) - g(x)) dx \\
 &= 2\pi \int_a^b x (f(x) - g(x)) dx \\
 &= 2\pi \bar{x} A \\
 &= (2\pi \bar{x}) A \\
 &= (\text{distance traveled by of the region}) \times (\text{the area of the region}).
 \end{aligned}$$